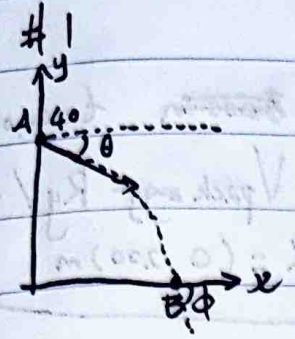
HW #5

#1



Given: $A(0.00, 40.00)\,m$, $B(B_x\hat{i}, 0.00)\,m$,
$\theta = 37^\circ$, $t = 2.00\,s$

$y = y_0 + V_{oy}t + \frac{1}{2}gt^2$

$0 = 40\,m + V_0 \sin 37^\circ \cdot 2.00\,s + \frac{1}{2} \cdot -9.81\,m/s^2 \cdot 2.00\,s^2$

$-40\,m = V_0 \sin 37^\circ \cdot 2.00\,s + -19.62\,m$

$-20\,m = V_0 \sin 37^\circ s - 9.81\,m$

$0 = V_0 \sin 37^\circ s + 10.19\,m$

$V_0 = 10.19\,m / 0.6018\,s$

$V_0 = 16.93\,m/s$

2)

$x = x_0 + V_{ox}t + \frac{1}{2}a_x t^2$

$= 0 + V_{ox}t + \frac{1}{2} \cdot 0 \cdot 2^2$

$= V_0 t$

$= 16.93\,m/s \cdot 2.00\,s$

$= 33.86\,m$

$= V_{ox}t$

$= 16.93\,m/s \cdot \cos 37^\circ \cdot 2.00\,s$

$= 16.93\,m/s \cdot .7986 \cdot 2.00\,s$

$= 27.04\,m$

b) $V_0 = .7986\hat{i} + .6018\hat{j}$

$V_x = V_{ox} + a_x t$

$= 16.93\,m/s \cdot \cos 37^\circ + 0 \cdot 2.00\,s$

$= 13.52$

$V_y = V_{oy} + a_y t$ $\quad a_y = g$

$= 16.93\,m/s \cdot \sin 37^\circ + -9.81\,m/s^2 \cdot 2.00\,s$

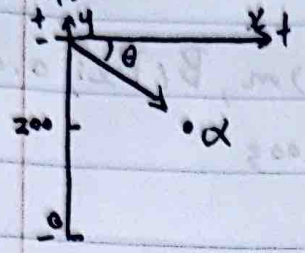
$= 10.19\,m/s - 19.62\,m/s$

$= -9.431\,m/s$

$\phi = \tan^{-1}\left(\frac{V_y}{V_x}\right)$

$= -34.89^\circ \approx -34.90^\circ$

#2



Given: $\theta = -31^\circ$, ~~R_{xy}~~ = ~~200m~~, $t = 4.00\,s$,

$V_{plane\,\alpha} = V_{pack\,0}$, $V_{pack\,angy} = R_y / t = 50\,m/s$

$\Delta y = -200\,m$, $\alpha = (0, 200)\,m$

~~$V_{pack\,0y} = gt - 50\,m/s$~~

~~$= -9.81\,m/s \cdot 4.00\,s - 50\,m/s$~~

~~$= -89.24\,m/s$~~

$\Delta y = V_{pack0y}\,t + \frac{1}{2} g t^2$

$V_{pack0y} = (\Delta y - \frac{1}{2} g t^2)/t$

$= \Delta y / t - \frac{1}{2} g t$

$= -\frac{200\,m}{4\,s} - \frac{1}{2} \cdot 9.81\,m/s^2 \cdot 4\,s$

$= -50\,m/s - 19.62\,m/s$

$= -69.62\,m/s$

2) $V_{plane\,\alpha} = -\frac{69.62\,m/s}{\sin 31^\circ}$

$= -135.9\,m/s$

$S_{plane\,\alpha} = |V_{plane\,\alpha}|$

$= 135.9\,m/s$

$V_{pack0y} = -69.62\,m/s$

$-69.62\,m/s = V_{pack0} \sin -31^\circ$

$+135.9\,m/s = V_{pack0}$

b)

set the x position where the package was dropped 0

$x = x_0 + V_0 t + \frac{1}{2} a t^2$

$= 0 + V_0 t + \frac{1}{2} \cdot 0 \cdot 4^2$

$= V_0 t$

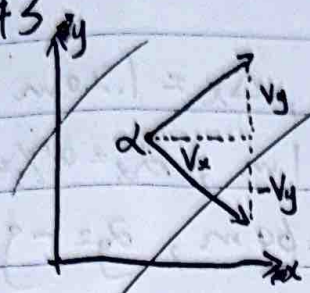
$x = V_{0x} t$

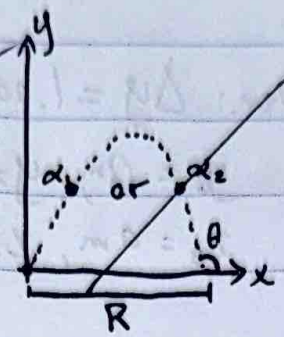
$= 135.9 \cdot \cos 31^\circ \cdot 4$

$= 135.9 \cdot .8572 \cdot 4$

$= 465.9739\,m$

#3





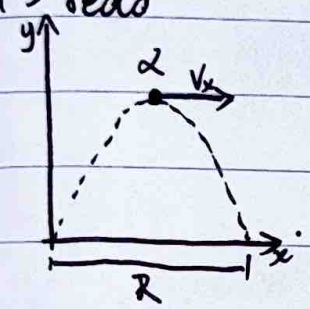
Given: at point α, $V_x = 15\ m/s$

$\alpha s = 3.00\ s$.

$y_0 = 0\ m$

$\vec{V} =$

#3 redo



Given: $V\alpha = 15\ m/s\ \hat{i} + 0\ m/s\ \hat{j}$, $\alpha t = 3.00\ s$

$V_x = \sqrt{15^2 + 0^2} = 15\ m/s$, $t = 2\alpha t$

a) $\cancel{V_y = V_{oy} + gt}$

$\cancel{= 0\ m/s + -9.81\ m/s^2 \cdot 3\ s}$

$\cancel{= -}$

$V_{oy} = -gt + V_y$

$= -9.81\ m/s^2 \cdot 3\ s + 0\ m/s$

$= 29.43\ m/s$

$V_0 = \sqrt{V_x^2 + V_{oy}^2}$

$= \sqrt{15^2 + 29.43^2}$

$= 33.03\ m/s$

b) $V_y = V_{oy} + gt$

$= 29.43\ m/s + -9.81\ m/s^2 \cdot 6\ s$

$= -29.43\ m/s$

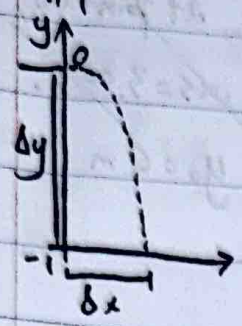
$\theta = \tan^{-1}\left(\frac{V_y}{V_x}\right)$

$= \tan^{-1}\left(-\frac{29.43}{15.00}\right)$

$= -62.99^\circ$

$R = V_x t = 90\ m$

#4



Given: $\Delta y = 1.00\,m$, $\Delta x = 1.20\,m$,
$y_0 = 0\,m$, $y_2 = -1\,m$, $a_x = 0\,m/s^2$,
$x_0 = 0\,m$, $x_2 = 1.60\,m$, $a_y = -g$,

a) $\cancel{y = y_0 + V_{0y}t + \frac{1}{2}a_y t^2}$

$\cancel{-V_{0y}t = \frac{1}{2}a_y t^2 + y_0 - y}$

$\cancel{V_{0y} = \frac{1}{2}a_y t + \frac{y_0 - y}{t}}$

$\cancel{= -\frac{1}{2}\cdot 9.81\,m/s}$

$\cancel{x = x_0 + V_{0x}t + \frac{1}{2}a_x t^2}$

$\cancel{V_{0x} = \frac{1}{2}a_x t + \frac{x - x_0}{t}}$

$\cancel{V_x^2 = V_{0x}^2 + 2a\left(\frac{x - x_0}{t}\right)}$

// V_{0y} should be $0\,m/s$ because it started by rolling on the platform.

b) $y_2 = y_0 + V_0 t + \frac{1}{2}a_y t^2$

$y_2 - y_0 = V_0 t + \frac{1}{2}a_y t^2$

$2(y_2 - y_0) = 2V_0 t + a_y t^2$

$2\cdot(-1-0) = 2\cdot 0\cdot t + -9.81\,m/s^2\cdot t^2$

$-2 = -9.81\,m/s^2\cdot t^2$

$t = \sqrt{.2039}\,s$

$t = 0.4515\,s$

a) $x_2 = x_0 + V_{0x}t + \frac{1}{2}a_x t^2$

$V_{0x}t = x_2 - x_0 - \frac{1}{2}a_x t^2$

$V_{0x} = \frac{x_2 - x_0}{t} - \frac{1}{2}a_x t$

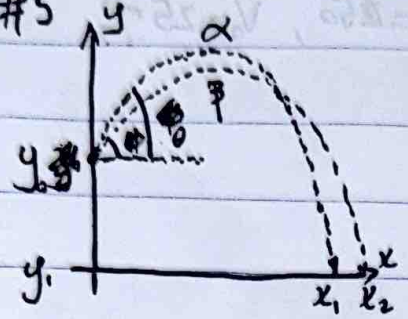
$= \frac{1.60m - 0m}{0.4515s} - \frac{1}{2}\cdot 0\,m/s^2\cdot 0.4515s$

$= \frac{1.60m}{.4515s}$

$= 3.544\,m/s$

$S_0 = |3.544\,m/s\,\hat{i} + 0.0\,m/s\,\hat{j}|$

#5



Given: $y_0 = 0\,m$, $y_1 = -2.100\,m$,

$x_0 = 0\,m$, $x_1 = 17.00\,m$, $V_{\alpha 0} = V_{\beta 0}$

$\phi = 40^\circ$, $\theta = 42^\circ$

$\cancel{x_1 = x_0 + V_{x0} t +}$

$y_1 = y_0 + V_{0y} t + \frac{1}{2} a t^2$

$V_{0y} t = y_1 - y_0 + g t^2 \cdot \frac{1}{2}$

$V_0 \sin\theta\, t = y_1 + g t^2 \cdot \frac{1}{2}$

$t = \frac{-V_0 \sin\theta \pm \sqrt{V_0 \sin^2\theta - 2 y_1 g}}{g}$

$x_1 = V_{0x}\, t$

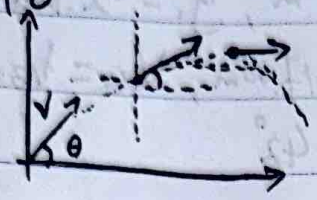
$x_1 = -V_0 \cos\theta \left(\frac{V_0 \sin\theta \pm \sqrt{V_0 \sin^2\theta - 2 g y_1}}{g} \right)$

$17\,m = -V_0 \cos 42^\circ \left(\frac{V_0 \sin 42^\circ + \sqrt{V_0 \sin^2 42^\circ - 2 \cdot 9.81\,m/s^2 \cdot 2.100\,m}}{9.81\,m/s^2} \right)$

$17\,m = -V_0 \cos 42^\circ \left(\frac{V_0 \sin 42^\circ + \sqrt{V_0 \sin^2 42^\circ - 41.202}}{9.81\,m/s^2} \right)$

$=$

#6



Given: $\theta = 50^\circ$, $V_0 = 25\ m/s$

~~$V = V_x i + V$~~

$V_x = V_{ox} = V_0 \cos\theta$

$= 25\ m/s \cdot \cos 50^\circ$

$= 16.069\ m/s$

$V_y = V_{oy} - gt = V_0 \sin\theta - gt$

$= 25\ m/s \cdot \sin 50^\circ - 9.81\ m/s^2\, t$

$= 19.151\ m/s - 9.81\ m/s^2\, t$

$\tan 30^\circ = V_y / V_x$

$.577 = \frac{19.151\ m/s - 9.81\ m/s^2\, t}{16.069\ m/s}$

$9.272\ m/s = 19.151\ m/s - 9.81\ m/s^2\, t$

$9.879\ m/s = 9.81\ m/s^2\, t$

$t = \frac{9.871\ m/s}{9.81\ m/s^2}$

$t = 1.00\ s$

#7

$$R = \frac{2V_0^2 \cos^2\theta(2+\tan\theta)}{g}$$

$y_0 = 2R$

$y = 0$

$x = R$

$x = V_{0x} t = V_0 \cos\theta \cdot t$

$y = V_{0y} t + at^2 + y_0$

$0 = 2R + V_0 \sin\theta \cdot t - \frac{1}{2} g t^2$

by quadradic equation

$$t = \frac{-V_0 \sin\theta + \sqrt{(V_0 \sin\theta)^2 + 4Rg}}{g}$$

$x = V_0 \cos\theta \, t$

$$R = V_0 \cos\theta \cdot \frac{-V_0 \sin\theta + \sqrt{(V_0 \sin\theta)^2 + 4Rg}}{g}$$

$$R = \frac{2V_0^2 \cos^2\theta(2+\tan\theta)}{g}$$

#8

$y_0 = y$

a) $R = y_{max}$

$V_y = V_{oy} + a_y t$

$a_y t = V_y - V_{oy}$

$t = \frac{V_y - V_{oy}}{g}$

$\because V_y$ is horizontal

$\therefore V_y = 0\ m/s$

$t_{max} = \frac{V_{oy}}{g}$

$= \frac{V_0 \sin\theta}{g}$

$y_{max} = V_{oy} t_{max} - g t_{max}^2 \cdot \frac{1}{2}$

$= \frac{(V_0 \sin\theta)^2}{g} - g\left(\frac{V_0 \sin\theta}{g}\right)^2 \frac{1}{2}$

$= \frac{(V_0 \sin\theta)^2}{g} - \frac{(V_0 \sin\theta)^2}{2g}$

$= \frac{(V_0 \sin\theta)^2}{g} - \frac{1}{2}\frac{(V_0 \sin\theta)^2}{g}$

$= \frac{(V_0 \sin\theta)^2}{2g}$

$t_{final} = 2 t_{max}$

$= \frac{2 V_0 \sin\theta}{g}$

$V_x = V_{ox} = V_0 \cos\theta$

$R = x_{final} = V_{ox} t_{final}$

$= V_0 \cos\theta \cdot \frac{2 V_0 \sin\theta}{g}$

$= \frac{2 V_0^2 \cos\theta \sin\theta}{g}$

$= \frac{V_0^2 \sin 2\theta}{g}$

$x_{final} = y_{max}$

$\frac{V_0^2 \sin 2\theta}{g} = \frac{(V_0 \sin\theta)^2}{2g}$

$V_0^2 \sin 2\theta = \frac{V_0^2 \sin^2\theta}{2}$

$\sin 2\theta = \frac{\sin^2\theta}{2}$

$2 \sin 2\theta = \sin^2\theta$

$2 \cdot 2 \cos\theta \sin\theta = \sin^2\theta$

$4 \cos\theta = \sin\theta$

$4 = \tan\theta$

$\theta = \tan^{-1}(4)$

$= 75.96°$

b) $R = \frac{1}{2} y_{max}$

$y_{max} = \frac{V_0^2 \sin^2\theta}{2g}$ $\qquad R = \frac{V_0^2 \sin 2\theta}{g}$

$R = \frac{1}{2} y_{max}$

$\frac{V_0^2 \sin 2\theta}{g} = \frac{1}{2} \cdot \frac{V_0^2 \sin^2\theta}{2g}$

$V_0^2 \sin 2\theta = \frac{V_0^2 \sin^2\theta}{4}$

$4 V_0^2 \sin 2\theta = V_0^2 \sin^2\theta$

$4 \cdot 2 \cos\theta \sin\theta = \sin^2\theta$

$8 \cos\theta = \sin\theta$

$8 = \tan\theta$

$\theta = \tan^{-1}(8)$

$= 82.87°$